Syngamy

Report: a2e44d8e
Generated: 2026-07-31 04:33:27Z
Run started: not recorded
Software: OriginMarker 2.2.1 (Kinetochore), Syngamy module
Sperm donor: GSM4472397_sperm_DNA_71.CEL.txt.gz (825,656 markers, SHA-256 1b56911dae7030f4)
Oocyte donor: none supplied, so a maternal origin is inferred from paternal absence rather than measured
Donor heterozygosity: 16.658% autosomal, of called markers
Samples: 1 analysed (1 gynogenetic)
Assembly: GRCh37
Data handling: every file was read in the browser that produced this report. No genotype was transmitted, and none is retained.

Result summary

Sample	Origin	Absence	Ceiling	x ceiling	Zygosity	Sperm	Informative
GSM4472418_donor_B_77.CEL.txt.gz	gynogenetic	5.25%	0.98%	5.4x	diploid	-	622,806

Absence is the rate at which the donor's obligate allele is missing where he is homozygous. Ceiling is what this sample's own noise can manufacture: its no-call rate times its heterozygous fraction. A sample is called as lacking a paternal contribution only above 3x that ceiling, and as carrying one only at or below it; in between it is left uncalled. No confidence percentage is reported, deliberately: see Interpretation.

GSM4472418_donor_B_77.CEL.txt.gz vs GSM4472397_sperm_DNA_71.CEL.txt.gz

GYNOGENETIC

No paternal contribution. Maternal in origin.

absent 5.25% | ceiling 0.98% | 5.4x | 622,806 informative markers | zygosity diploid

Evidence

Measurement	Observed	Reference	The reference is
Donor alleles absent	5.25%	0.98%	this sample's no-call rate times its heterozygous fraction. Dropout manufactures absence only by turning a heterozygous call homozygous and discarding the donor's allele, so the bound is the product and neither term alone.
Alleles the donor lacks	16.13%	8.33%	half the donor's heterozygosity. A second parent supplies an allele he lacks only where he is homozygous, at sum(pq), and his heterozygosity is sum(2pq).
Heterozygous BAF band	16.27%	8.00%	measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.

Sex chromosomes

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chrX	19,277	2,351	12.20%	12.5	absent	

Chromosomes

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chr1	51,292	2,576	5.02%	5.1	absent	
chr2	48,581	2,331	4.80%	4.9	absent	
chr3	41,339	1,883	4.56%	4.7	absent	
chr4	37,328	1,853	4.96%	5.1	absent	
chr5	36,186	1,828	5.05%	5.2	absent	
chr6	43,254	2,806	6.49%	6.6	absent	
chr7	33,468	1,493	4.46%	4.6	absent	
chr8	30,789	1,652	5.37%	5.5	absent	
chr9	26,959	1,391	5.16%	5.3	absent	
chrX	19,277	2,351	12.20%	12.5	absent	
chr10	30,505	1,735	5.69%	5.8	absent	
chr11	33,455	1,958	5.85%	6.0	absent	
chr12	29,883	1,527	5.11%	5.2	absent	
chr13	20,542	1,089	5.30%	5.4	absent	
chr14	20,951	1,133	5.41%	5.5	absent	
chr15	19,465	1,246	6.40%	6.5	absent	
chr16	22,973	1,223	5.32%	5.4	absent	
chr17	22,970	1,136	4.95%	5.1	absent	
chr18	17,311	1,037	5.99%	6.1	absent	
chr19	21,539	1,066	4.95%	5.1	absent	
chr20	15,214	720	4.73%	4.8	absent	
chr21	8,872	587	6.62%	6.8	absent	
chr22	9,930	451	4.54%	4.6	absent	

Sample quality

Markers: 825,656 read, 801,414 called (97.1%)
No-call: 2.94%
Heterozygous: 16.66% of called

Research use only. Parent of origin here is a statistical call from array genotypes and requires confirmation by an independent method in a qualified genetics laboratory. Not a clinical diagnostic.

chrX / autosomal het:	1.176
Sex call:	female
Product:	UK Biobank Axiom Array
Assembly:	GRCh37 (0 placements illegal under GRCh37, 6,315 under GRCh38, 825,656 tested)
Allele coding:	ok BAF corroborates 0=AA, 2=BB
Mean BAF AA/AB/BB:	0.028 / 0.503 / 0.976
SHA-256:	c0854c0daedac886ca41124459af071519ff4dc588e2872e64cac7572ddda133
File size:	62,719,288 bytes

Quality gates

Gate	Value	Verdict	Basis
call rate	97.1%	usable	inside the vendor bulk-DNA band
het-to-hom asymmetry valid	97.1%	usable	holds within the usable call-rate band
genome-wide LOH	16.7%	report only	no genome-wide LOH signature
heterozygosity plausible	16.7%	usable	within the plausible range for one diploid genome
numeric genotype coding	-	usable	BAF corroborates 0=AA, 2=BB
sex call	117.6%	usable	chrX/autosome heterozygosity ratio 1.176 is inside the female band. Dropout cancels in the ratio, so this survives amplified material.

Findings

No contribution from this parent anywhere. chrY alone would not have separated this from an X-bearing sperm carrying a full paternal genome.

Methods

Parent of origin was determined from SNP array genotypes using OriginMarker 2.2.1 (Kinetochore). For each of 1 sample(s), the rate at which the sperm donor's obligate allele was absent was computed across autosomal markers where he is homozygous and the sample is called, and compared against the rate that sample's own genotyping noise can produce, taken as the product of its no-call rate and its heterozygous fraction. A sample was called as lacking a paternal contribution only where the observed rate exceeded that bound by 3-fold, and as carrying one only where it fell at or below the bound; rates between the two were left uncalled. Whether a second parent contributed was assessed from the rate at which the sample carries alleles the sperm donor does not possess, against half his heterozygosity, which is the contribution a second parent makes. Zygosity was taken from the fraction of B-allele frequencies falling in the heterozygous band, 0.35 to 0.65. Sperm type was inferred from the chrX rate rather than from chrY, so that an X-bearing sperm carrying a full paternal genome is distinguishable from an absent paternal contribution; neither case leaves a Y.

No oocyte donor array was supplied. A maternal origin below is therefore inferred from the absence of the paternal contribution rather than measured, and a sample belonging to neither declared parent is indistinguishable from one that is maternal in origin.

Assembly was determined from marker positions against the UCSC chromInfo and gap tables and found to be GRCh37. No liftOver was performed.

Interpretation

No confidence percentage is attached to a call, and this is deliberate. The per-sample likelihood ratio between "paternal genome present" and "absent" runs past 10^10000 on a genome-wide array, which is not a probability any reader should be handed. The honest figure is the empirical accuracy of the method, and on the pairs of known relationship it has been run against that is 9 of 9 correct, whose 95% lower bound is roughly 72%. About 300 consecutive correct calls would be needed to claim 99%. What is reported instead is the margin: the observed rate, the reference it was measured against, and the ratio between them, so a reader can see how far from the boundary the call sits.

The absence axis separates by roughly thirty-fold and carries the call. The alleles-the-donor-lacks axis separates by about 1.6-fold and only distinguishes a biparental genome from a paternal-only one that is heterozygous; treat a near-boundary call on that axis as provisional. A sample scoring in the uncalled band is not a weak positive, it is an absence of evidence: an array from the second parent measures dropout directly and settles it.

Constants and where they come from

Constant	Value	Provenance
Mendelian inconsistency floor	0.0063	Kothiyal 2019, 0.63% of variant sites across 1,314 nuclear families. A lower bound on the spurious-violation rate, never the value itself.
Paternal-absence noise bound	no-call rate x heterozygous fraction	Derived per sample, not fitted. Bounds every related pair measured, within 2x: 0.13% predicted against 0.05% observed, 0.18% against 0.16%, 10.4% against 9.69%.
Absence calling margin	3x	How far above the noise bound an absence must sit before it is called.
Residual absence floor	0.005	Absence on clean data from genotyping error alone, measured at 0.03% and 0.05%. Added to the bound so a near-perfect array still has a non-zero reference.
Second-parent signal	half the donor's heterozygosity	Derived. A second parent supplies an allele the donor lacks only where he is homozygous, at sum(pq); his heterozygosity is sum(2pq) over the same markers.
Diploid heterozygous BAF band	0.08 of markers in 0.35-0.65	Measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.
Per-chromosome minimum	200 informative markers	Below this a chromosome is not reported: the rate is too noisy to place against the ceiling.
Assembly tables	UCSC chromInfo and gap, hg19 and hg38	Freely redistributable. No liftOver chain file is used or shipped: those carry a non-commercial field-of-use restriction Apache 2.0 cannot sublicense.

Files read

File	Role	Bytes	Markers	SHA-256
GSM4472397_sperm_DNA_71.CEL.txt.gz	donor	63,674,062	825,656	1b56911dae7030f4ecebc2ade9c7b1bacee23def81d130e6b3d0ac9edb388169
GSM4472418_donor_B_77.CEL.txt.gz	sample	62,719,288	825,656	c0854c0daedac886ca41124459af071519ff4dc588e2872e64cac7572ddda133

The hash is of the file as read, so a reviewer can confirm the input without being sent it.

Citation

OriginMarker 2.2.1 (Kinetochore). Syngamy: parent of origin from SNP array genotypes. Report a2e44d8e, generated 2026-07-31 04:33:27Z.
Kothiyal P et al. Mendelian inconsistent signatures from 1,314 ancestrally diverse family trios. J Comput Biol 2019;26(5):405-419. doi:10.1089/cmb.2018.0253. Cited for the inconsistency floor above.